Chris Law

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Skills

Languages: C++, C#, Java, Python, JavaScript, HTML/CSS, Verilog, SQL
Technologies: Altium, React, PostgreSQL, Crow, REST APIs, Arduino, Git, Figma, Wix Studio

Education

University of Waterloo — BASc, Computer Engineering Sept 2025 – Apr 2030
- Currently hold a 3.84/4.00 GPA and the President's Scholarship of Distinction

Work Experience

UX/UI & SEO Analyst — Real Estate Institute of Canada May 2026 – Aug 2026
- Evaluated five webpages for UX/UI and SEO as part of Waterloo's WE Accelerate program
- Simplified course discovery for three career-stage audiences by redesigning key pages in Wix Studio
- Developed SEO recommendations using keyword-focused headings, concise metadata, and internal links
- Presented research-backed design and content recommendations directly to REIC leadership

Web Development Co-op — Payment Source Apr 2024
- Built responsive PaySimply interfaces by translating Figma mockups into HTML and CSS
- Improved mobile usability by adapting layouts and interface components across screen sizes
- Improved access to customer support by implementing and integrating a help-centre widget
- Delivered reviewed UI updates through Git branches, pull requests, and team documentation

Produce Clerk — Real Canadian Superstore 2022 – 2025
- Maintained product quality and availability while serving customers in a fast-paced department
- Supported efficient restocking by coordinating incoming pallets with receiving staff
- Onboarded new employees by explaining procedures, answering questions, and providing guidance

Projects

Trading Simulator — C++20, PostgreSQL, React, Crow
- Built a full-stack trading simulator supporting user accounts, live pricing, and portfolio tracking
- Displayed live prices and historical candlestick data by integrating the Alpaca Markets API
- Preserved transaction consistency by implementing atomic trades with balance and share validation
- Supported portfolio tracking by modelling users, holdings, and trades in indexed PostgreSQL tables
- Protected credentials and database operations using password hashing and parameterized SQL queries

ESP32-S3 Development Board — Altium Designer
- Translated ESP32-S3, USB-C, USB-UART, and regulator datasheets into a four-layer PCB design
- Implemented native USB and USB-C-to-UART programming with dual ports and an FT231X bridge
- Applied USB requirements using 5.1 kΩ CC pulldowns and 27 Ω series termination
- Designed 3.3V power and return paths with TL1963A regulation, internal planes, and thermal vias
- Resolved boot-strapping and octal-PSRAM pin restrictions before routing GPIO breakout headers

Terraria Content Mod — C#, Terraria API, Procedural Generation
- Reached over 700 lifetime unique subscribers by publishing a Terraria mod on Steam Workshop
- Expanded gameplay progression by adding custom items, ores, recipes, and progression content
- Added a procedurally generated biome while maintaining compatibility with existing world systems

Flappy Bird AI — Python, NEAT-Python
- Evolved neural networks to play Flappy Bird autonomously using the NEAT algorithm
- Improved obstacle navigation across generations using survival-based fitness rewards
- Visualized training progress and generational performance trends using Matplotlib

Automated CAM Assessment — Arduino, C++
- Automated a randomized 13-question cognitive assessment using an Arduino-based prototype
- Enabled audio capture and playback using a microphone, speaker, OLED, and microSD storage
- Prepared recorded responses for Whisper transcription and CAM-based assessment scoring